

CFD@HUST

Comprehensive User Manual & Technical Documentation

1. GENERAL INTRODUCTION

The **CFD@HUST** is an automated Graphical User Interface (GUI) designed to simplify complex Aerodynamics and Hydrodynamics simulations using the OpenFOAM solver engine. It abstracts complex Linux terminal commands, meshing dictionaries, and solver setups into a streamlined Windows interface via WSL (Windows Subsystem for Linux).

The software is divided into three highly specialized physical modules:

- **Wind Tunnel Module:** Dedicated to external, incompressible aerodynamics (e.g., Automotive, Architectural, Drones).
- **Towing Tank Module:** Dedicated to free-surface, multiphase hydrodynamics (e.g., Ship hulls, Floating offshore structures).
- **Airfoil Module:** Dedicated to compressible, transonic/supersonic aerodynamics for 2D wing profiles and blades.

2. UNIVERSAL WORKFLOW (ALL MODULES)

Regardless of the module you are using, the core simulation pipeline dictates that you follow these exact steps sequentially:

1. **New Project:** Click "New Project" on the toolbar and select an entirely empty folder. This initializes an isolated OpenFOAM case directory. *Warning: Do not use spaces or special characters in the folder path.*
2. **Import STL Geometry:** Browse and select your 3D CAD model (.stl format). Ensure the model is a closed, watertight manifold surface.
3. **Configure Units & Rotation:** Select the physical unit your CAD software exported in (Meters, Millimeters, Inches). The software will automatically scale the vertices to Meters. Use the Rotation (Z) value to ensure the front of your model directly faces the incoming flow (which always travels from -X to +X).

4. **Setup Case:** Click the "Setup Case" button. The software writes all OpenFOAM dictionaries (controlDict, fvSchemes, boundary conditions) into the **0**, **constant**, and **system** folders.
5. **Generate Mesh:** Click "Generate Mesh" to execute the **blockMesh** and **snappyHexMesh** processes. Once finished, toggle "Show Mesh" to inspect the cell quality in the 3D viewer.
6. **Run Solver:** Specify the number of CPU Cores and click "RUN". The mesh will be decomposed and the parallel solver will commence.
7. **Reconstruct Results:** *Crucial step.* If you stop the simulation early, or once it naturally finishes, you must click "Reconstruct Results". This merges the fragmented **processor** folders into single time-step directories so the data can be read by ParaView.

3. DETAILED MODULE GUIDE

3.1 Wind Tunnel Module

This module utilizes the **simpleFoam** solver for steady-state, incompressible, turbulent flows.

- **Domain Scaling:** The bounding box (virtual wind tunnel) is defined using multipliers of your object's physical dimensions (L = Length, W = Width, H = Height).
 - **Front Scale:** Distance upstream. Recommended: 2.0 to 3.0 L.
 - **Back Scale:** Distance downstream (wake region). Recommended: 5.0 to 7.0 L to allow pressure recovery without boundary reflection.
 - **Side / Top Scale:** Distance to tunnel walls. Recommended: 3.0 to 5.0 to avoid artificial blockage effects which artificially increase drag.
- **Mesh Generation:** Use the built-in "Y+ Calculator" to determine the **First Layer Thickness** based on the inlet velocity and target Y+ value (typically ~30 for Wall Functions). Set Refinement Level to 4 or 5 for accurate drag predictions.
- **Physics:** Define the inlet velocity in m/s. For bluff bodies like cars, the **kOmegaSST** turbulence model is highly recommended over **kEpsilon** due to superior flow separation prediction.

3.2 Towing Tank Module

This module utilizes the **interFoam** solver for transient, multiphase (air and water) flow utilizing Volume of Fluid (VOF) phase tracking.

- **Domain Dimensions:** You must input the absolute Tank Length, Width, and Height directly in meters, rather than using multipliers.
- **Water Level (Z-axis):** This is the most critical parameter. It sets the global Z-coordinate for the calm water free-surface interface. Ensure your ship's origin ($Z=0$) aligns correctly with this water level to achieve the intended physical draft.
- **Time Step (Delta T):** Multiphase solvers are notoriously unstable. You must set a very small initial Delta T (e.g., 0.001s) to ensure the Courant (CFL) number remains below 1.0. The solver is configured to automatically adjust the time step after initialization.
- **Simulation Time:** Set the End Time long enough to allow the vessel to travel at least 2 to 3 ship lengths. This allows the Kelvin wave wake pattern to fully develop.

3.3 Airfoil Module

This module utilizes compressible solvers (e.g., **rhoCentralFoam** or **shockFluid**) tailored for transonic and supersonic aerospace profiles.

- **Geometry & Angle of Attack (AoA):**
 - **Chord Length:** Input the exact aerodynamic chord of your wing in meters.
 - **Angle of Attack:** Input the angle in degrees. The software automatically resolves the freestream velocity vector into U_x and U_y components. *Do not manually rotate the STL file to simulate AoA.*
- **Physics (Mach Number):** Velocity is defined strictly by the **Mach** number (e.g., 0.85) instead of m/s. The solver internally computes the actual freestream velocity based on standard atmospheric conditions (Temperature = 300K, Pressure = 101325 Pa).
- **Meshing:** This module automatically generates a pseudo-2D domain (1 cell thick in the Z-direction) using **mirrorMesh** or **extrudeMesh** techniques. This ensures perfectly symmetric 2D flow and drastically reduces computation time for cross-sectional analysis.

4. MONITORING AND VISUALIZATION

- **Live Residuals:** Monitor the "Residuals" tab. For steady-state (Wind Tunnel), residuals should drop steadily below $1e-4$. For transient solvers (Towing Tank / Airfoil), residuals will spike and drop at every time step.
- **Live Forces:** The "Results" tab plots the Drag (C_d) and Lift (C_l) coefficients dynamically. The simulation is considered converged when these lines flatten out horizontally.

- **ParaView Automation:** Clicking the "ParaView" button executes a Python macro. It automatically loads your reconstructed data, generates center-plane slices, applies color mapping for velocity/pressure, and sets up streamlines without requiring manual configuration.

5. AI ASSISTANT & UTILITIES

- **Gemini Integration:** Paste your Gemini API key in the right panel. The AI is prompt-engineered specifically for OpenFOAM diagnostics.
 - **Analyze Error:** If the OpenFOAM console outputs a "Fatal Error" or "Floating Point Exception", click this button. The AI will parse the terminal log to identify the root cause (e.g., "Courant number exceeded 1.0", "Mesh skewness too high") and suggest a precise fix.
 - **Suggest Settings:** Click this to have the AI calculate estimated Reynolds numbers, suggest boundary conditions for k and omega/epsilon, and recommend domain sizes based on your current inputs.
 - **Export CSV:** Saves the raw aerodynamic force convergence history into a standard .CSV file for external processing in Excel, Python, or MATLAB.
 - **Clean Case:** A utility that securely deletes heavy processor directories and mesh files to free up disk space while preserving your configuration files (sim_config.json) for future runs.
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